Cboe Global Indices

Cboe PutWrite Indices

METHODOLOGY

© Cboe Global Indices, LLC. All rights reserved.

Contents

1 Introduction 2
1.1 Index Objective 2
1.2 Supporting Documentation 2
2 Index Construction 3
2.1 Index Constituents 3
2.2 Index Rebalance 3
2.2.1. Rebalance Details PUTY & PUTR 4
2.2.2. Rebalance Details PTLT 4
2.2.3. Rebalance Details WPTR & WPUT 5
2.3 Options Pricing 5
3 Index Calculations 6
3.1 Non-Roll Date Calculations 6
3.1.1. Non-Roll Date Calculations PUTR, PUTY, WPTR & WPUT 6
3.1.2. Non-Roll Date Calculations PTLT 7
3.2 Roll Date Calculations 7
3.2.1. Roll Date Calculations PUTR & PUTY 7
3.2.2. Roll Date Calculations PTLT 8
3.2.3. Roll Date Calculations WPTR & WPUT 9
3.3 Intraday Calculations 10
4 Calculation and Dissemination 10
5 Index Information 11
6 Appendix 1 - Changes 11
7 Appendix 2 - Document Information 11

1 Introduction

This document provides a transparent and easily accessible view of the methodology used to calculate select Cboe PutWrite Indices, referred to in this document as "the Indices".

1.1 Index Objective

The index objectives of the Indices covered by this document are as follows:

- The **Cboe Russell 2000 PutWrite Index (PUTR)** is designed to track the performance of a hypothetical passive investment strategy that collects option premiums from writing an At-the-Money (ATM) RUT Put option on a monthly basis and holds a Treasury bill account invested in one-month Treasury bills to cover the liability from the short RUT Put option position.

- The **Cboe S&P 500 2% OTM PutWrite Index (PUTY)** is designed to measure and track the performance of a hypothetical passive investment strategy that collects option contract premiums from writing a 2% Out-of-the-Money ("OTM") SPX put option on a monthly basis and holds a Treasury bill account invested in one-month T-bills to cover any liability from the short SPX Put option position.

- The **Cboe 20+ Year Treasury Bond PutWrite Index (PTLT)** is designed to track the performance of a hypothetical passive investment strategy that collects option premiums from writing an At-the-Money (ATM) TLT Put option on a monthly basis and holds a Treasury bill account invested in one-month Treasury bills to cover the liability from the short TLT Put option position.

- The **Cboe Russell 2000 One-Week PutWrite Index (WPTR)** is designed to track the performance of a hypothetical passive investment strategy that collects option premiums from writing an At-the-Money (ATM) RUT Put option on a weekly basis and holds a Treasury bill account invested in one-month Treasury bills to cover the liability from the short RUT Put option position.

- The **Cboe S&P 500 One-Week PutWrite Index (WPUT)** is designed to track the performance of a hypothetical passive investment strategy that collects option premiums from writing an At-the-Money (ATM) SPX Put option on a weekly basis and holds a Treasury bill account invested in one-month Treasury bills to cover the liability from the short SPX Put option position.

1.2 Supporting Documentation

This Methodology should be read in conjunction with the following document:

Cboe_Index_Policies_Practices

2 Index Construction

2.1 Index Constituents

The Indices consist of the components as listed below.

Index	Type	Position	Ticker	Underlying Name	Expiration	Option Type
Cboe Russell 2000 PutWrite Index (PUTR)	T-bills	Long	n/a	n/a	4 weeks	n/a
	Option	Short	RUT	Russell 2000	1 month	Put
Cboe S&P 500 2% OTM PutWrite Index (PUTY)	T-bills	Long	n/a	n/a	4 weeks	n/a
	Option	Short	SPX	S&P 500	1 month	Put
Cboe 20+ Year Treasury Bond PutWrite Index (PTLT)	T-bills	Long	n/a	n/a	4 weeks	n/a
	Option	Short	TLT	iShares 20+ Year Treasury Bond ETF	1 month	Put
Cboe Russell 2000 One-Week PutWrite Index (WPTR)	T-bills	Long	n/a	n/a	4 weeks	n/a
	Option	Short	RUT	Russell 2000	1 week	Put
Cboe S&P 500 One-Week PutWrite Index (WPUT)	T-bills	Long	n/a	n/a	4 weeks	n/a
	Option	Short	SPX	S&P 500	1 week	Put

2.2 Index Rebalance

Rebalances are performed according to the following table.

Index Name	Index Ticker	Roll Date	Strike Determination	Strike Price Selection (K)
Cboe Russell 2000 PutWrite Index	PUTR	Roll date is the third Friday of each month. Should the third Friday fall on an exchange holiday, the roll date is the preceding business day.	Roll Date	The first available strike below the last disseminated value of the Russell 2000 Index before 11:00 a.m. ET.
Cboe S&P 500 2% OTM PutWrite Index	PUTY	Roll date is the third Friday of each month. Should the third Friday fall on an exchange holiday, the roll date is the preceding business day.	Roll Date	The first available strike below the 98% of the last disseminated value of the S&P 500 Index before 11:00 a.m. ET.
Cboe 20+ Year Treasury Bond PutWrite Index	PTLT	Roll date is the third Friday of each month. Should the third Friday fall on an exchange holiday, the roll date is the preceding business day.	Roll Date	The first available strike lower than or equal to the last disseminated value of the TLT ETF before 4:00 p.m. ET.
Cboe Russell 2000 One-Week PutWrite Index	WPTR	Roll date is every Friday. Should an exchange holiday fall on a Friday, the roll date is the preceding business day.	Roll Date	The first available RUT Put option strike below the Special Opening Quotation (SOQ) of the Russell 2000 Index.

Cboe S&P 500 One-Week PutWrite Index	WPUT	Roll date is every Friday. Should an exchange holiday fall on a Friday, the roll date is the preceding business day.	Roll Date	The first available SPX Put option strike below the Special Opening Quotation (SOQ) of the S&P 500 Index (SPX Index)

2.2.1. Rebalance Details PUTY & PUTR

On the initial roll date of the Indices, a monthly put option is written. At the same time, a notional amount equal to the strike (K) of the put option is invested in a treasury bill account to cover the liability of the short put option position.

The Indices require that the put options in the hypothetical portfolio be held to maturity. The Special Opening Quotation ("SOQ") of the underlying index is used as the final settlement price of the index put options. The SOQ is a special calculation of the underlying index that is compiled from the opening prices of component stocks underlying the index. The SOQ calculation is performed when all the constituents underlying the equity index have opened for trading and is usually determined before 11:00 a.m. ET. The final settlement price of the put option at maturity is the greater of 0 and the strike price of the expiring put option minus the SOQ.

Subsequent to the settlement of the expiring put option, a new put option expiring in the next month is then deemed written, or sold, a transaction commonly referred to as a "roll." The option is selected in line with the rules described in in the table above. Simultaneously, a notional amount equal to the strike (K) of the put option is invested in a Treasury bill account. The Treasury bill account will accumulate interest at the 4-week Bank discount rate.

Once the strike price of the new put option has been identified, the new put option is deemed sold at a price equal to the volume-weighted average of the traded prices ("VWAP") of the new put option between 11:30 a.m. and 12:00 p.m. ET. Cboe Global Indices uses the transaction prices and trade codes from the Options Price Reporting Authority (OPRA) feed to calculate the VWAP. The VWAP is calculated in two steps:

1. Orders categorized by OPRA as uppercase A – H and lowercase f – t in the new Put option that occur during the VWAP period are excluded to remove late, cancel, and spread orders.
2. The weighted average for all remaining transaction prices of the new Put option during the VWAP period is calculated, with weights equal to the fraction of total remaining volume transacted at each price during this period. If no transactions occur in the new put option during the VWAP period, then the new put option is deemed sold at the last bid price reported before the end of the VWAP period.

2.2.2. Rebalance Details PTLT

On January 21, 2005, the initial roll date of the PTLT Index, an ATM monthly TLT Put option is written. The strike price of the ATM TLT Put option is the first available strike lower than or equal to the last disseminated value of the TLT ETF before 4:00 p.m. ET. The premium collected from writing the TLT option is the last NBBO bid price of the day on the written option. Typically, on the third Friday (Roll Day) of every month since the initial roll date, the TLT Put option settles and subsequently, a new monthly ATM TLT Put is written. Following the same rule as the initial roll date, the strike price of the ATM TLT Put option is the first available strike lower than or equal to the last disseminated value of the TLT ETF before 4:00 p.m. ET. The premium collected from writing the TLT Put option is the last NBBO bid price of the day on the written option. The Treasury bill account will accumulate interest at the 4-week Bank discount rate.

2.2.3. Rebalance Details WPTR & WPUT

On the initial roll date of the Indices, an ATM one week put option is written after the market opens (9:30 a.m. ET). The strike is selected in line with the rules described in the table above. The first bid quote after the market opens is the option premium collected from the written put option. At the same time, a notional amount equal to the strike (K) of the put option is invested in a Treasury bill account to cover the liability of the short put option position. These actions are repeated on every roll date since the initial roll date. The Treasury bill account will accumulate interest at the 4-week Bank discount rate.

As the expiring put option can be either an AM-settlement or PM-settlement on the Roll Date, this would affect the settlement procedure of the expiring put option. If the expiring put option is AM-settlement on the Roll Date, the expiring put option settles against the SOQ. If the expiring put option is PM-settlement on the Roll Date, the expiring put option is purchased back at the last ask price of the put option before 4:00 p.m. ET. For AM-settlement, the first available put option strike below the SOQ is selected, and the option premium collected is the first bid quote of the put option after 9:30 a.m. ET. For PM-settlement, the first available put option strike below the last disseminated value of the underlying index before 4:00 p.m. ET is selected, and the option premium collected is the last bid quote of the put option before 4:00 p.m. ET. In either case, a Treasury bill account with initial cash equal to the strike (K) of the put option is set up simultaneously to cover the potential liability of the short put option position.

2.3 Options Pricing

The put option quotes are sourced from one or multiple Cboe Options Exchanges, via the Options Price Reporting Authority (OPRA) feed, as per the table below. For Indices where option prices are sourced from multiple Cboe Options Exchanges, the Best Bid and Offer prices are used.

Index Name	Index Ticker	Cboe Options Exchange	Cboe C2 Options Exchange
Cboe Russell 2000 PutWrite Index	PUTR	X	
Cboe S&P 500 2% OTM PutWrite Index	PUTY	X	
Cboe 20+ Year Treasury Bond PutWrite Index	PTLT	X	X
Cboe Russell 2000 One-Week PutWrite Index	WPTR	X	
Cboe S&P 500 One-Week PutWrite Index	WPUT	X	

3 Index Calculations

The Indices covered by this document, apart from PTLT, are calculated according to the following formula:

$$Index_t = Index_{t-1} * (1 + R_t)$$

where:

- $Index_t$ is the level of the Index;
- $Index_{t-1}$ is the level of the Index on the previous day; and
- R_t is the return of the Index.

3.1 Non-Roll Date Calculations

3.1.1. Non-Roll Date Calculations PUTR, PUTY, WPTR & WPUT

The non-roll date return of the index is calculated as:

$$(1 + R_t) = \frac{(M_t - Put_t)}{(M_{t-1} - Put_{t-1})}$$

$$R_f = \left(\frac{1}{1 - \frac{28}{360} * \text{USB4WTA}}\right)^{\frac{t-(t-1)}{28}}$$

$$M_t = (1 + R_f) * M_{t-1}$$

where:

- M_t and M_{t-1} are the values of the Treasury bill account on day *t* and *t-1,* respectively;
- R_f is the 4-Week Bank discount rate de-annualized to the daily rate. The 4-week Bank discount rate is obtained from the website of the U.S. Department of the Treasury;
- Put_t is the average of the last bid-ask quote of the Put option before 4:00 p.m. ET on date *t* for the closing value. For intraday calculations, the average of the current reported bid and ask prices of the put option is used; and
- Put_{t-1} is the average of the last bid-ask quote of the Put option before 4:00 p.m. ET on the previous day.

3.1.2. Non-Roll Date Calculations PTLT

The non-roll date return of the index is calculated as:

$$Index_t = M_t - N * P_t$$

$$M_t = (1 + R_f) * M_{t-1}$$

$$R_f = \left(\frac{1}{1 - \frac{28}{360} * \text{USB4WTA}} \right)^{\frac{t-(t-1)}{28}}$$

where:

- M_t and $M_{t-1}$are the values of the Treasury bill account on day *t* and *t-1,* respectively;
- R_f is the 4-Week Bank discount rate de-annualized to the daily rate. The 4-week Bank discount rate is obtained from the website of the U.S. Department of the Treasury;
- N is the unit of TLT Put options sold on the previous roll date; and
- P_t is the average of the last NBBO quote of the TLT Put option before 4:00 pm ET for the closing value. For intraday calculations, the average of the current reported NBBO prices of the put option is used.

3.2 Roll Date Calculations

3.2.1. Roll Date Calculations PUTR & PUTY

The roll date return of the index is calculated in two steps using the following formulas.

The following formula is used to calculate the return from the previous day's market close to present day's morning settlement:

$$(1 + R_1) = \frac{(M_old_{t-1} - Put_old_{settle})}{(M_old_{t-1} - Put_old_{t-1})}$$

where:

- M_old_{t-1} is the value of the Treasury bill account on the previous day;
- $Put_old_{settle} = Max(0, K_{old} - SOQ_t)$ is the settlement value of the expiring Put option, where K_{old} the strike of the expiring option; and
- Put_old_{t-1} is the average of the last bid-ask quote of the expiring Put option before 4:00 p.m. ET on the previous day.

The following formula is used to calculate the return from the moment the new Put option is deemed sold to market close:

$$(1 + R_2) = \frac{(M_new_t - Put_new_t)}{(M_new_t - Put_new_{vwap})}$$

where:

- M_new_t is the cash allocated to cover the potential loss of the new Put option on the roll day;
- Put_new_t is the average of the last bid-ask quote of the new Put option before 4:00 p.m. ET; and
- Put_new_{vwap} is the volume weighted average price ("VWAP") of the new Put option.

Note: interest is not accumulated in the Treasury bill account on the roll day.

The product of the two parts is the total return of the roll day:

$$(1 + R_t) = (1 + R_1) \times (1 + R_2)$$

3.2.2. Roll Date Calculations PTLT

$$Index_t = M_t - N_{new} * P_new_t$$

$$M_t = R_f * M_{t-1} - N_{old} * P_{settle} + N_{new} * P_new_{bid}$$

$$N_{new} = \frac{(Index_{t-1})}{(S_{t-1})}$$

where:

- P_new_t is the average of the last NBBO quote of the new Put option before 4:00 p.m. ET;
- N_{new} is the unit of TLT Put options sold on the roll date;
- N_{old} is the unit of the expiring TLT Put option;
- P_{settle} = Max (0, K_old - S_t) is the final settlement price of the expiring Put option, where K_old is the strike of the expiring Put option;
- P_new_{bid} is the last NBBO bid quote of the new Put option before 4:00 p.m.ET; and
- S_t *and* S_{t-1} is the closing price of TLT on day t and t-1, respectively.

3.2.3. Roll Date Calculations WPTR & WPUT

AM-settlement roll days

For AM-settlement roll days, the following formula is used to calculate the return from the previous day's market close to present day's morning settlement:

$$(1 + R_1) = \frac{(M_t - Put_old_{settle})}{(M_{t-1} \quad - Put_old_{t-1})}$$

where:

- $Put_old_{settle} = Max(0, K_{old} - SOQ_t)$ is the settlement value of the expiring put option;
- Put_old_{t-1} is the average of the last bid-ask quote of the expiring put option before 4:00 p.m. ET on the previous day.

The following formula is used to calculate the return from the moment the new put option is deemed sold to market close:

$$(1 + R_2) = \frac{(M_new_t - Put_new_t)}{(M_new_t - Put_new_{bid\ 930})}$$

where:

- M_new_t is the cash allocated to cover the potential loss of the new put option on the roll day;
- Put_new_t is the average of the last bid-ask quote of the new put option before 4:00 p.m. ET; and
- $Put_new_{bid\ 930}$ is the first bid quote of the new put option after 9:30 a.m. ET.

Note: interest is not accumulated in the Treasury bill account on roll day.

PM-settlement roll days

For PM-settlement roll days, the following formula is used to calculate the return from the previous day's market close to 4:00 p.m. ET:

$$(1 + R_1) = \frac{(M_{t-1} \quad - Put_old_{ask\ 4pm})}{(M_{t-1} - Put_old_{t-1})}$$

where:

- $Put_old_{ask\ 4pm}$ is the last ask quote of the expiring put option before 4:00 p.m. ET;
- Put_old_{t-1} is the average of the last bid-ask quote of the expiring put option before 4:00 p.m. ET on the previous day.

The following formula is used to calculate the return from 4:00 p.m. ET to market close (4:15 p.m. ET):

$$(1 + R_2) = \frac{(M_new_t - Put_new_t)}{(M_new_t - Put_new_{bid\ 4pm})}$$

where:

- M_new_t is the cash allocated to cover the potential loss of the new put option on the roll day;
- Put_new_t is the average of the last bid-ask quote of the new put option before 4:00 p.m. ET; and
- $Put_new_{bid\ 4pm}$ is the last bid quote of the new put option before 4:00 p.m. ET.

Note: interest is not accumulated in the Treasury bill account on roll day.

The product of the two parts is the total return of the Roll Day:

$(1 + R_t) = (1 + R_1) * (1 + R_2)$

3.3 Intraday Calculations

On non-roll dates, the intraday calculations follow the non-roll date calculations.

On roll dates, the PUTR and PUTY Indices will not be disseminated intraday until after each indices' VWAP period has ended and the SOQ value has been made available. On roll dates, PTLT will not be disseminated intraday.

On a.m. settlement roll days, the WPUT and WPTR indices will not be disseminated intraday until the SOQ is determined.

4 Calculation and Dissemination

The Indices are calculated and disseminated as follows on each Business Day. A Business Day is defined as a day when the Cboe Options Exchanges are open for the Cboe Regular Trading Hours (RTH) session. The Indices follow the Cboe Options Exchange holiday schedule.

Index Name	Index Ticker	Dissemination Frequency	Dissemination Hours
Cboe Russell 2000 PutWrite Index	PUTR	15 seconds	9:30 a.m. – 4:00 p.m. ET
Cboe S&P 500 2% OTM PutWrite Index	PUTY	15 seconds	9:30 a.m. – 4:00 p.m. ET
Cboe 20+ Year Treasury Bond PutWrite Index	PTLT	15 seconds	9:30 a.m. – 4:00 p.m. ET
Cboe Russell 2000 One-Week PutWrite Index	WPTR	15 seconds	9:30 a.m. – 4:00 p.m. ET
Cboe S&P 500 One-Week PutWrite Index	WPUT	15 seconds	9:30 a.m. – 4:00 p.m. ET

5 Index Information

Index Name	Index Ticker	Base Date	Launch Date	Base Value	Currency
Cboe Russell 2000 PutWrite Index	PUTR	January 31, 2001	November 23, 2015	100	USD
Cboe S&P 500 2% OTM PutWrite Index	PUTY	June 30, 1986	February 15, 2019	100	USD
Cboe 20+ Year Treasury Bond PutWrite Index	PTLT	January 21, 2005	March 31, 2021	100	USD
Cboe Russell 2000 One-Week PutWrite Index	WPTR	February 28, 2011	November 23, 2015	100	USD
Cboe S&P 500 One-Week PutWrite Index	WPUT	January 31, 2006	August 3, 2015	100	USD

6 Appendix 1 - Changes

Major changes described in this document since October 14, 2024, are as follows:

Change Summary	Effective Date	Previous Language	Updated Language
-	-	-	-

7 Appendix 2 - Document Information

Version Number[1]	2.0
Last Revised Date	October 15, 2025

[1] *Prior to October 14, 2024, the methodologies of the Indices covered by this document were separately maintained in a legacy format.*

Disclosures and Disclaimers Related to Cboe Indices

Cboe Global Indices, LLC and Cboe Europe Indices, B.V. (collectively, the "Cboe Index Providers", and individually a "Cboe Index Provider"), are both subsidiaries of Cboe Global Markets, Inc. (Cboe Global Markets, Inc., the Cboe Index Providers, and the other subsidiaries and affiliates of Cboe Global Markets, Inc. are collectively referred to herein as the "Cboe Companies" and those entities are individually referred to as a "Cboe Company").

The Cboe Index Providers are each an index provider that administers, calculates and distributes indices and other values (as applicable) (collectively, "Cboe Indices" and individually a "Cboe Index"). Additional information regarding the Cboe Index Providers and their operations is available at **www.cboe.com/indices/**.

Nothing herein constitutes an offer of services. The information herein is provided solely for informational purposes. Cboe Indices are calculated and administered (as described in relevant methodologies, rules and other information) without regard to the needs of any particular person, entity, or group of persons, using market information, including, without limitation, market data from exchanges owned and operated by other Cboe Companies.

Index and benchmark values for dates or time periods prior to an index launch date, if any, are calculated using a theoretical approach involving back-testing historical data in accordance with the methodology in place on the launch date (unless otherwise stated). Back-testing involves the theoretical application of the index methodology in hindsight and therefore does not involve actual performance and may not produce performance commensurate with prospective application of the methodology. No back-tested approach can completely account for the impact of decisions that might have been made if calculations were made at the same time as certain underlying market conditions occurred. Supporting documentation for statements, comparisons, statistics or other technical data provided in this document is available by request at **www.cboe.com/contact/**.

Past performance of an index or financial product is not indicative of future results. Indices are not financial products that can be invested in directly, but they can be used as the basis for financial products (for example, without limitation, options, futures, mutual funds or exchange-traded funds) or to help manage portfolios.

The Cboe Index Providers are not investment advisers or tax advisers. No decision to buy, hold, or sell any financial product should be made in reliance on any of the information provided herein, and no representation is made regarding the advisability or tax consequences of any investment, whether based on an index or not. Nothing herein should be construed as investment advice. No representation or assurance is made that any financial product will accurately track the performance of an underlying index or provide positive investment returns. Index values and returns are theoretical, and do not reflect the results of actual trading of financial products, nor the value of an actual portfolio, and may not reflect the fees, taxes or other expenses associated with buying, holding or selling investments or the effect of actual market conditions or events. The Cboe Index Providers are not broker-dealers, intermediaries, investment firms or securities exchanges and they do not manage actual financial products. None of the Cboe Companies sponsor, endorse, promote, manage, sell or administer any financial product that is created, sponsored, issued or sold by a third-party, whether based on any Cboe Index or not.

No data, values, or other content contained herein ("Content") may be copied, modified, reverse-engineered, reproduced, distributed, or stored in a database or retrieval system, in any form or by any means, nor may any Content be used to create a derivative work (e.g., a financial product, service or index) or used to verify or correct other data or information, without the Cboe Index Provider's prior written consent. Without derogating from the aforementioned, Content shall not be used for any unlawful or unauthorized purposes.

The Cboe Companies, their third-party service or data providers, or any party from whom they have licensed trademarks or indices (collectively, the "Cboe Parties") do not guarantee the accuracy, completeness, or timeliness of the Content, trademarks, strategies or values, or the methodologies or input data used to calculate index values. The Cboe Parties are not responsible for any errors or omissions, regardless of the cause, or for the results obtained from the use of the Content, index strategies or values, methodologies or input data. THE CONTENT, INDICES AND RELATED STRATEGIES AND METHODOLOGIES ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, WITHOUT LIMITATION, ANY WARRANTY WITH RESPECT TO MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE. USERS OF CONTENT AND/OR THE CBOE INDICES BEAR ALL RISKS OF LOSS.

WITHOUT LIMITING THE FOREGOING AND TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, IN NO EVENT SHALL THE CBOE PARTIES BE RESPONSIBLE OR LIABLE TO ANY PERSON OR ENTITY FOR ANY DAMAGES OF ANY KIND, INCLUDING, BUT NOT LIMITED TO, DIRECT, INDIRECT, SPECIAL, INCIDENTAL, CONSEQUENTIAL, PUNITIVE, LOST PROFITS OR LOST OPPORTUNITIES, WHETHER OR NOT A CBOE PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING LIMITATIONS SHALL APPLY REGARDLESS OF WHETHER A CLAIM ARISES IN CONTRACT, TORT, NEGLIGENCE, STRICT LIABILITY, OR OTHERWISE.

The Cboe Index Providers receive compensation in connection with licensing Cboe Indices to third parties and providing calculation services to third parties. Some Cboe Companies operate markets for the trading of financial products and provide services to many organizations, including issuers of securities, investment advisers, broker-dealers, investment banks, other financial institutions and financial intermediaries, and accordingly may provide or receive fees or other economic benefits to or from those organizations. These organizations may also be Cboe Index Provider licensees or customers, and the Cboe Index Providers may include securities of or relating to such organizations in Cboe Indices.

Cboe®, C2SM, Cboe ClearSM, Cboe Closing Cross®, Cboe DigitalSM, Cboe LIS®, Cboe Futures ExchangeSM, Cboe Global IndicesSM, Cboe Global Markets®, Cboe Options Institute®, Cboe Volatility Index®, Bats®, BIDS Trading®, Buffer ProtectSM, BuyWriteSM, BYX®, BZX®, CFE®, EDGA®, EDGX®, Eris Exchange®, ErisX®, f(t)optionsSM, FLEX®, FLexible EXchange®, HanweckSM, Hybrid®, LiveVol®, NANO®, Options Institute®, PutWriteSM, Silexx®, SRVIXSM, Trade AlertSM, VIX®, and XSPSM are trademarks and service marks of the Cboe Companies. Trademarks of the Cboe Companies may only be used with written permission, subject to applicable Cboe Company trademark usage guidelines as published from time to time. Third-party trademarks and service marks used in this document are described below. The values and methodologies for indices branded with the "Cboe" trademark are the property of the relevant Cboe Index Provider.

The Cboe Companies have licensed several trademarks from third parties from whom they have also licensed indices which may be included in this document or in other materials related to Cboe Indices, as follows:

- Dow Jones®, Dow Jones Industrial Average® and DJIA® are trademarks or service marks of Dow Jones Trademark Holdings LLC ("Dow Jones"), have been licensed for use for certain purposes by the Cboe Companies. Derivative indices created, used and distributed by Cboe Companies and any investment products based thereon are not sponsored, endorsed, sold or promoted by Dow Jones, and Dow Jones makes no representation regarding the advisability of investing in any investment product that is based on any such derivative indices.
- FTSE® and the FTSE indices are trademarks and service marks of FTSE International Limited, used under license.
- IHS, IHS Markit, CDX, iBoxx, and iTraxx are trademarks of IHS Markit Limited and its affiliates ("IHS") and have been licensed for use by the Cboe Companies. Any IHS Markit index referenced herein is the property of IHS Markit and is used under license. Any products that have an IHS Markit index as their underlying interest are not sponsored, endorsed, or promoted by IHS Markit.
- MSCI and the MSCI index names are service marks of MSCI Inc. ("MSCI") or its affiliates and have been licensed for use by the Cboe Companies. Any derivative indexes and any financial products based on the derivative indexes ("Products") are not sponsored, guaranteed or endorsed by MSCI, its affiliates or any other party involved in, or related to, making or compiling such MSCI index. Neither MSCI, its affiliates nor any other party involved in, or related to, making or compiling any MSCI index makes any representations regarding the advisability of investing in such Products; or any warranty, express or implied; or bears any liability as to the results to be obtained by any person or any entity from the use of any such MSCI index or any data included therein. No purchaser, seller or holder of any Product, or any other person or entity, should use or refer to any MSCI trade name, trademark or service mark to sponsor, endorse, market or promote Products without first contacting MSCI to determine whether MSCI's permission is required.
- Nasdaq®, Nasdaq-100®, and Nasdaq-100 Index®, are trademarks of Nasdaq, Inc. (together with its affiliates, the "Corporations") and are licensed for use by Cboe Exchange, Inc. The CBOE Nasdaq-100 Volatility Index (the "Volatility Index") and CBOE Nasdaq-100 BuyWrite Index (the "BuyWrite Index") are not derived, maintained, published, calculated or disseminated by the Corporations. Neither the Volatility Index, the BuyWrite Index nor any product based on such indexes have been passed on by the Corporations as to their legality or suitability. Such products are not issued, endorsed, sold, or promoted by the Corporations. THE CORPORATIONS MAKE NO WARRANTIES AND BEAR NO LIABILITY WITH RESPECT TO THE VOLATILITY INDEX OR BUYWRITE INDEX.
- Russell, Russell 1000®, Russell 2000®, Russell 3000® and Russell MidCap® are registered trademarks of the Frank Russell Company, used under license.
- S&P, S&P 500, US500, 500, THE 500, SPDR, SPX, and DSPX are trade names or trademarks of S&P Dow Jones Indices, LLC or its affiliates ("S&P"). Any products that have an S&P index as their underlying interest are not sponsored, endorsed, sold or promoted by S&P and S&P makes no representation regarding the advisability of investing in any investment product that is based on any such indices.

© 2025 Cboe Exchange, Inc. and its affiliates. All Rights Reserved.

v20230405

Cboe Global Indices

CONTACT US

IndexSupport@cboe.com
www.cboe.com/indices